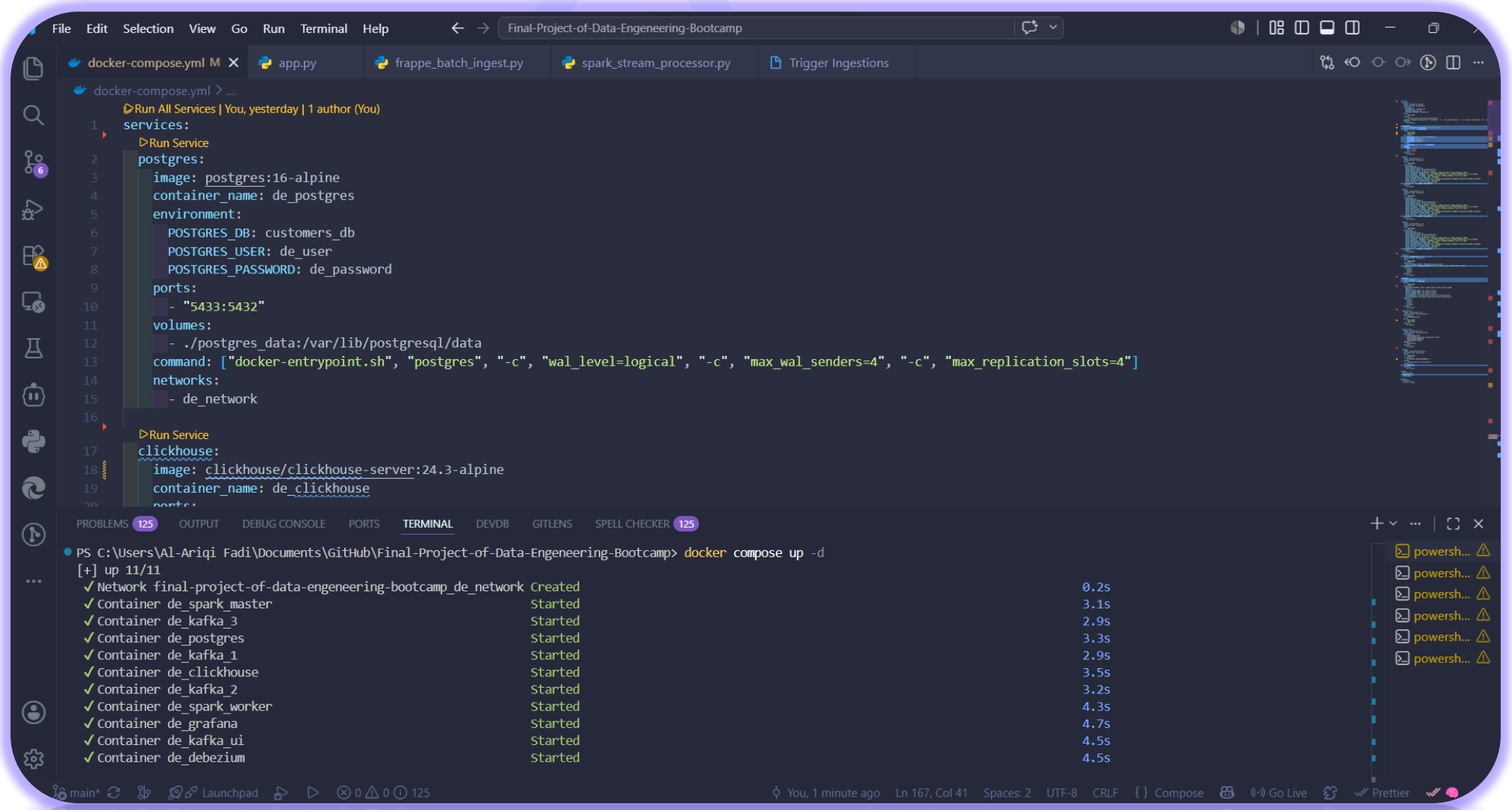
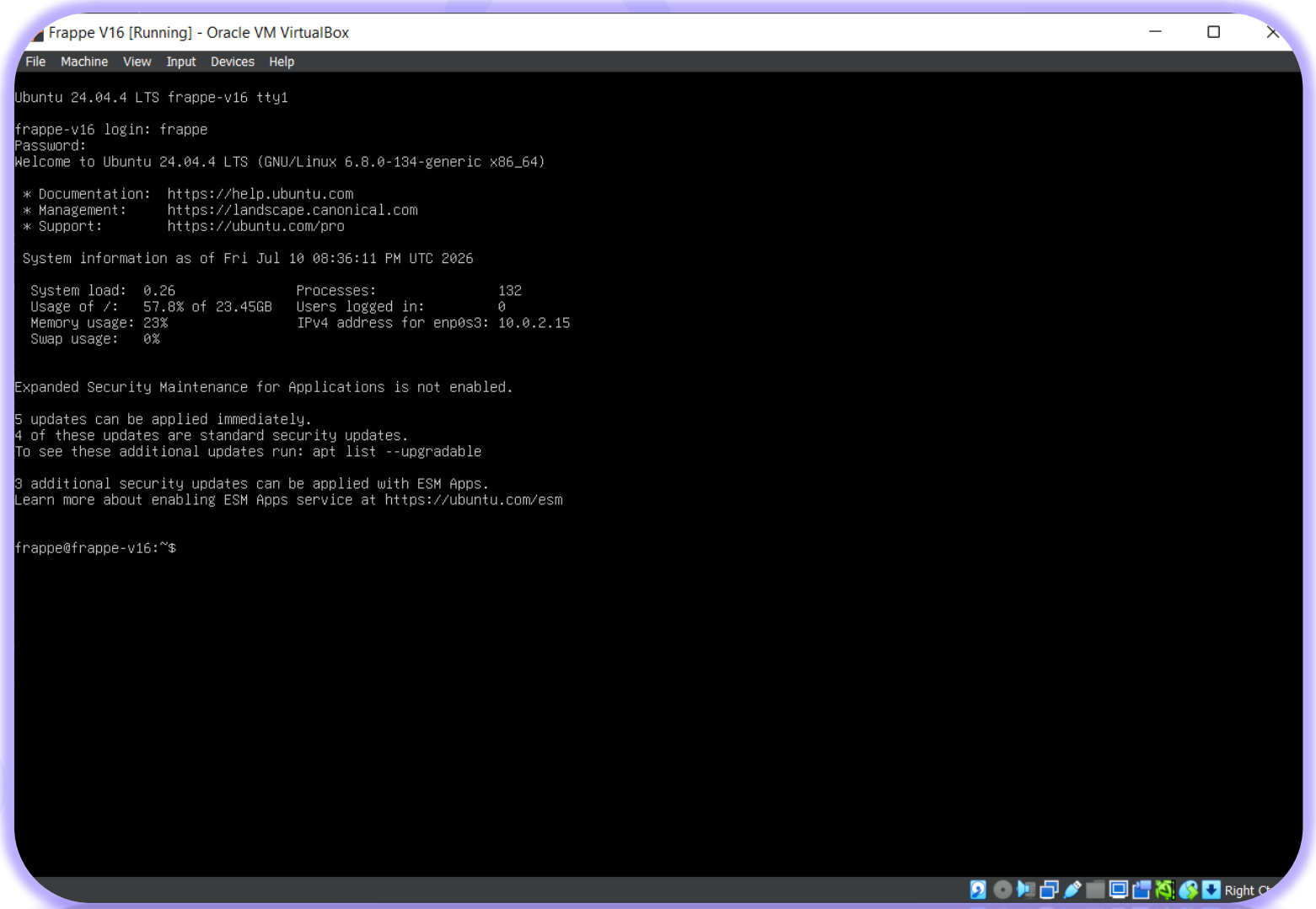
Appendix: Project Implementation Screenshots & System Demonstration

Eng. Fadi Al-Ariqi

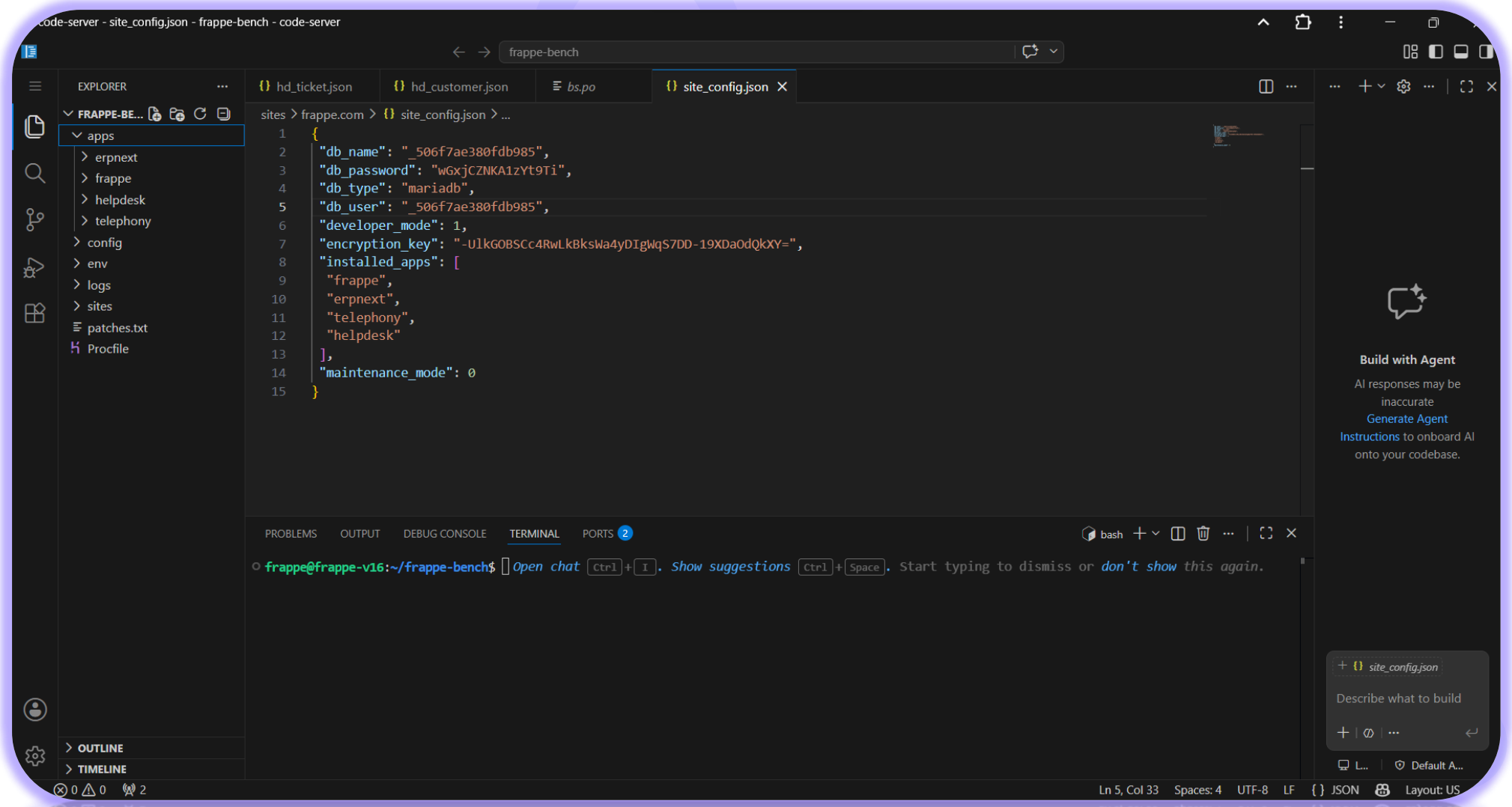
Starting the System Docker Containers



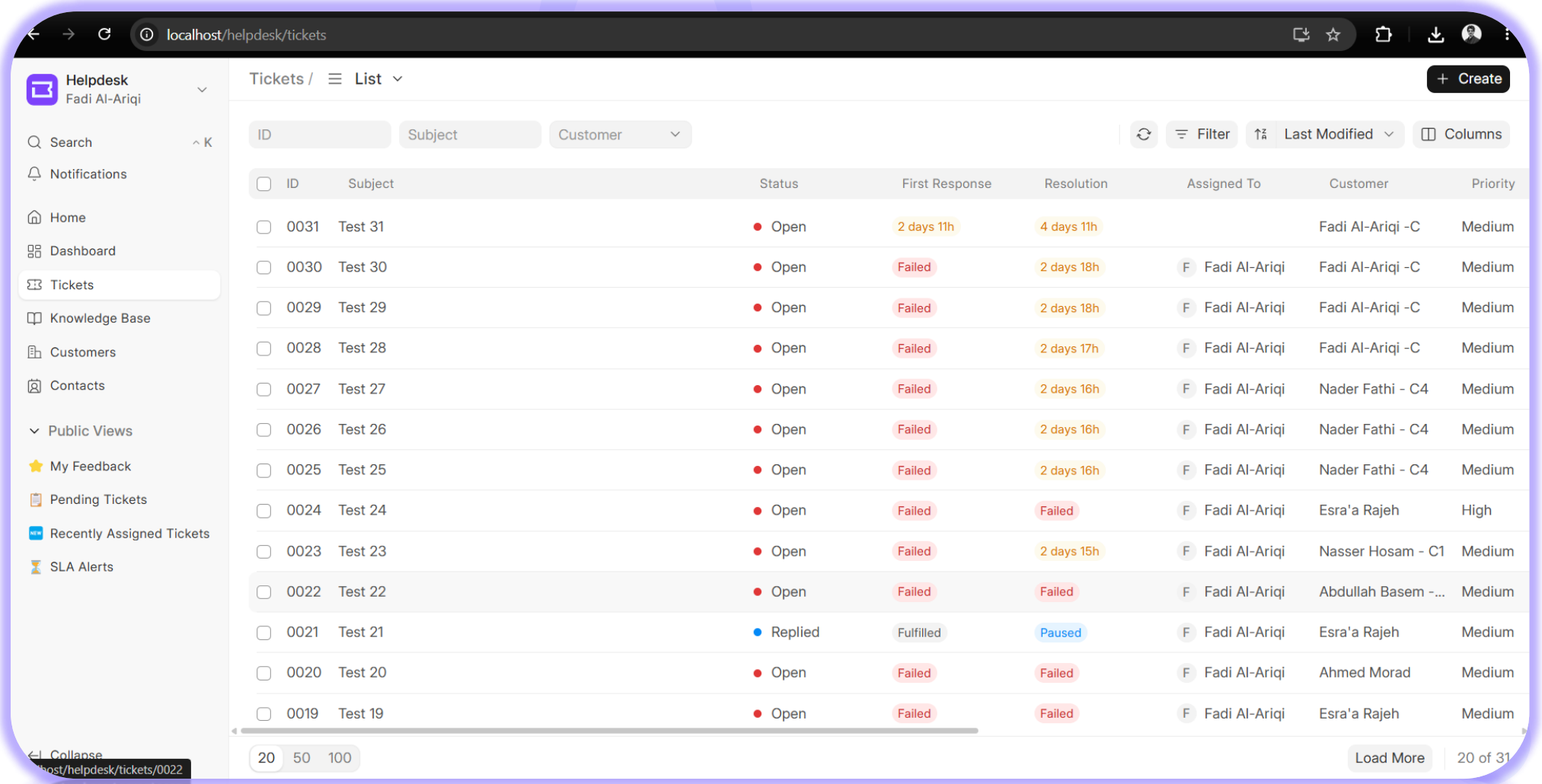
Starting the Frappe Application Server



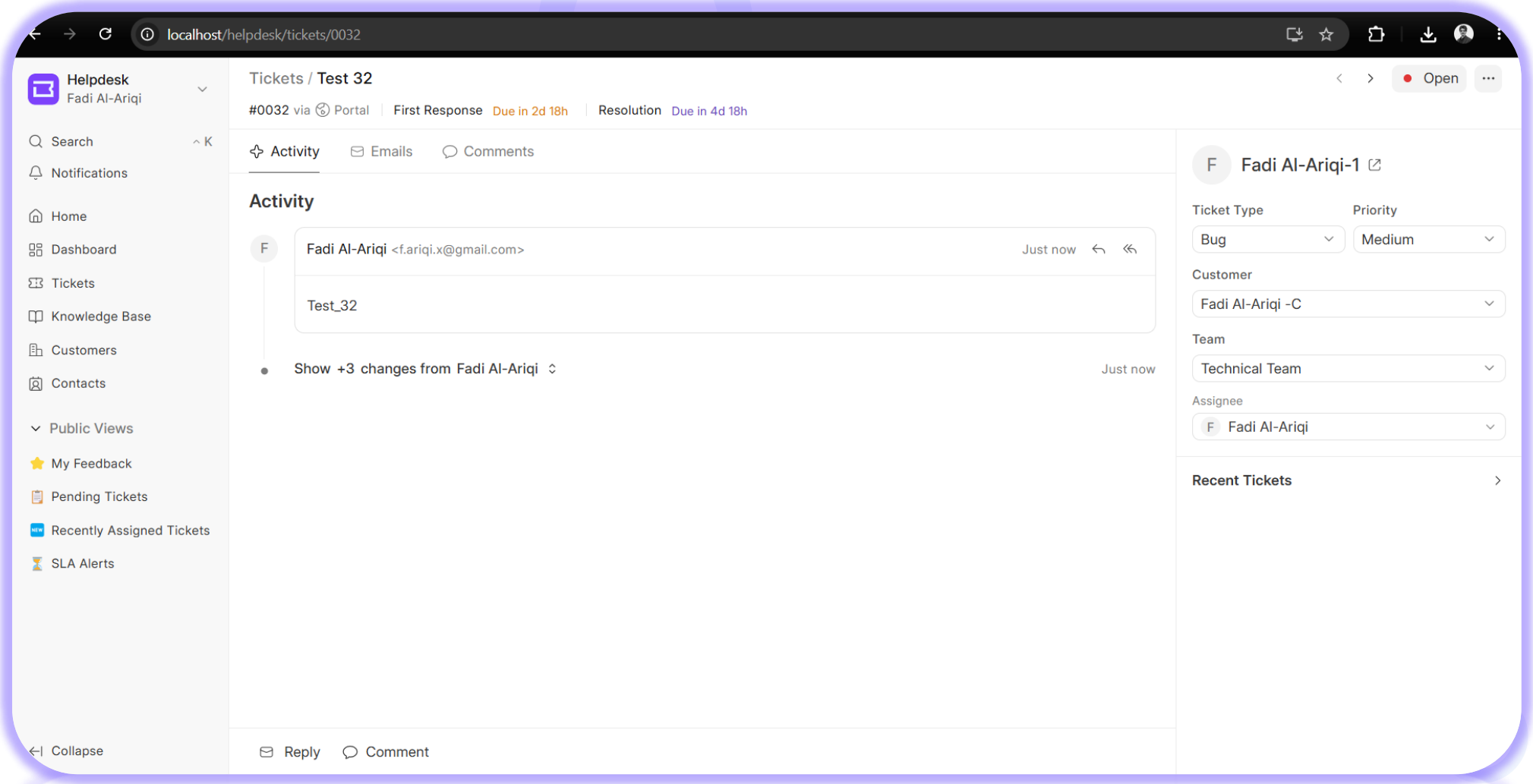
Verifying Successful System Deployment and Access



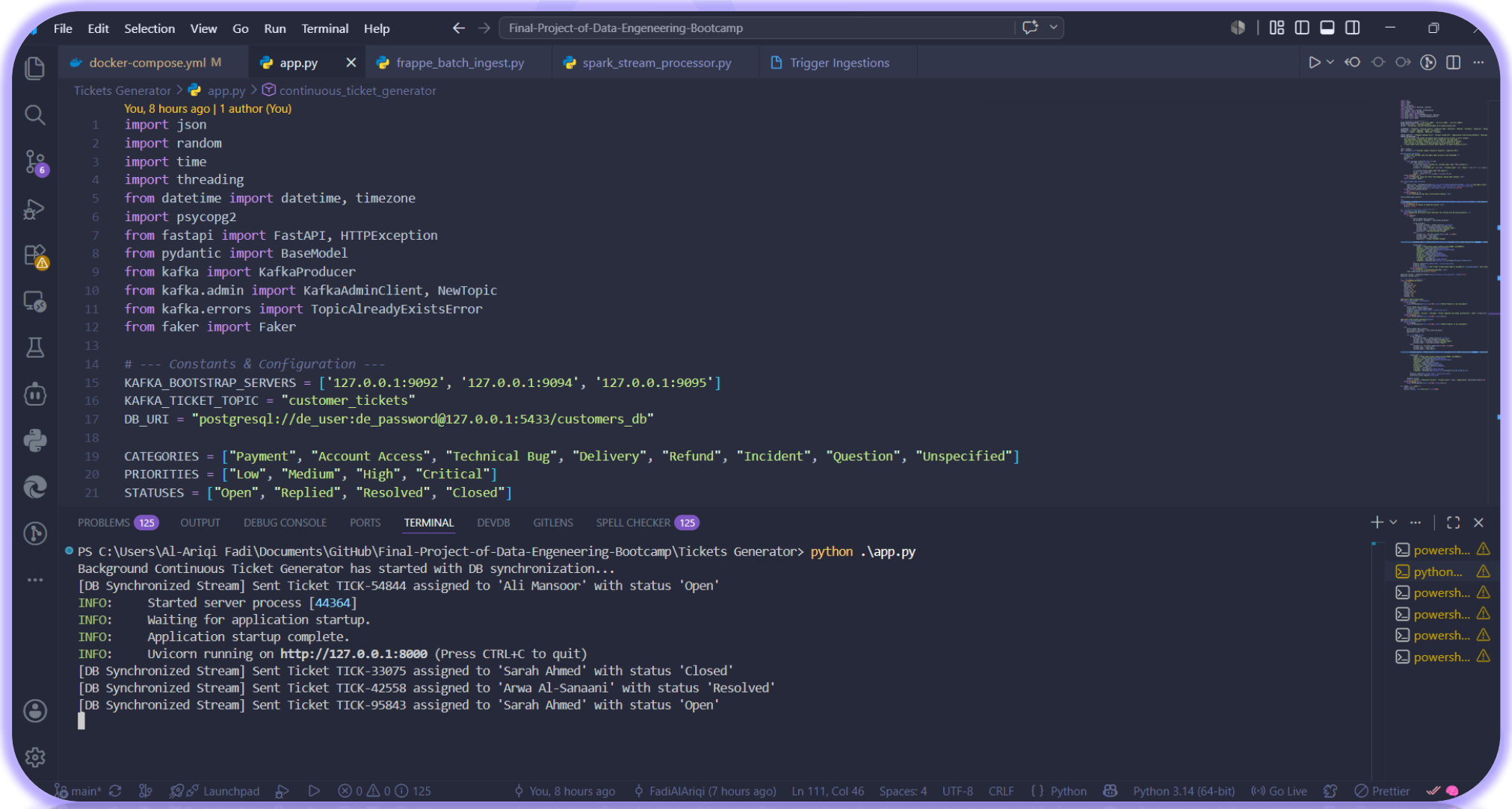
Frappe Helpdesk – Ticket Management Interface



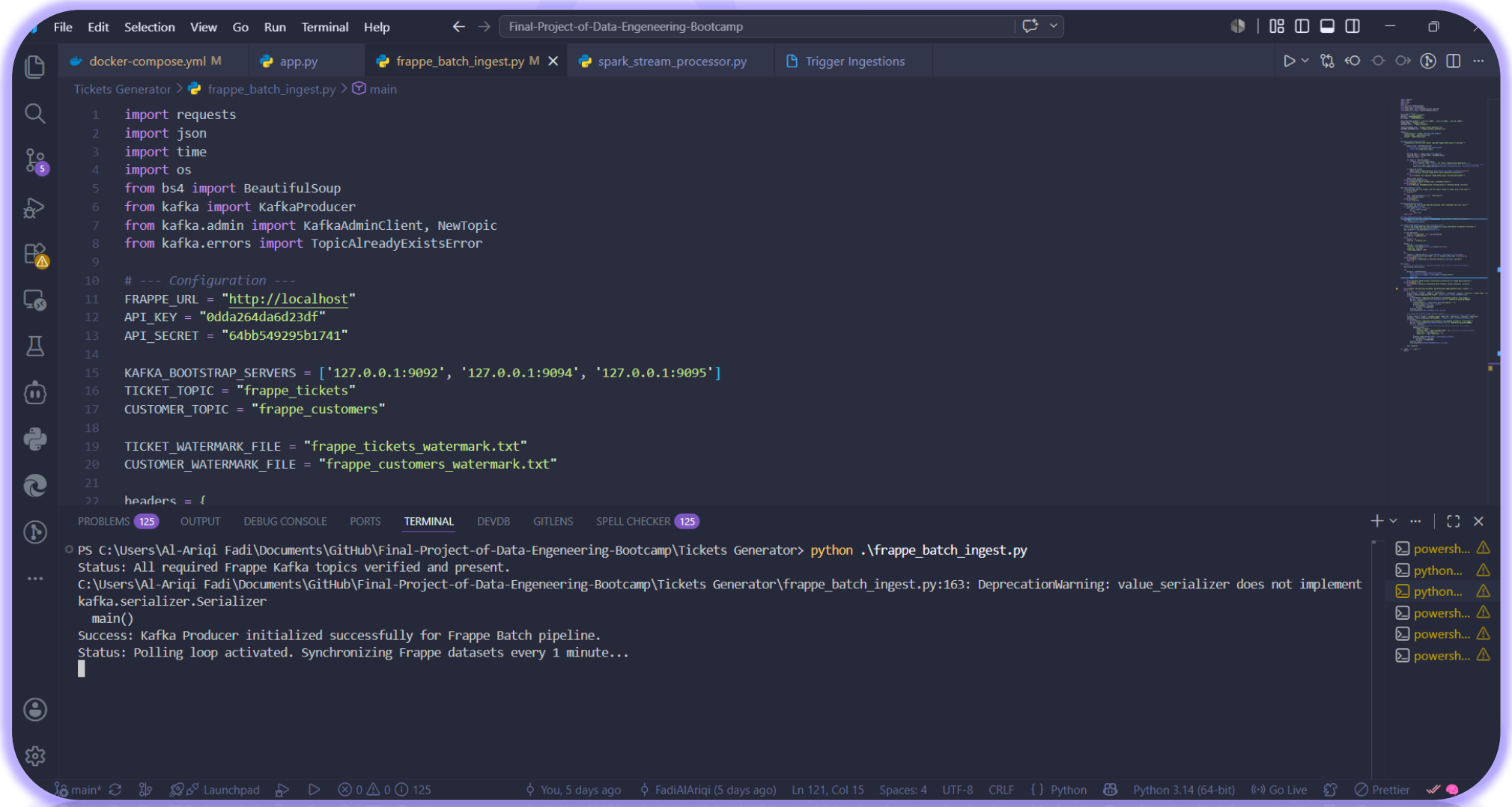
Support Ticket Creation Interface



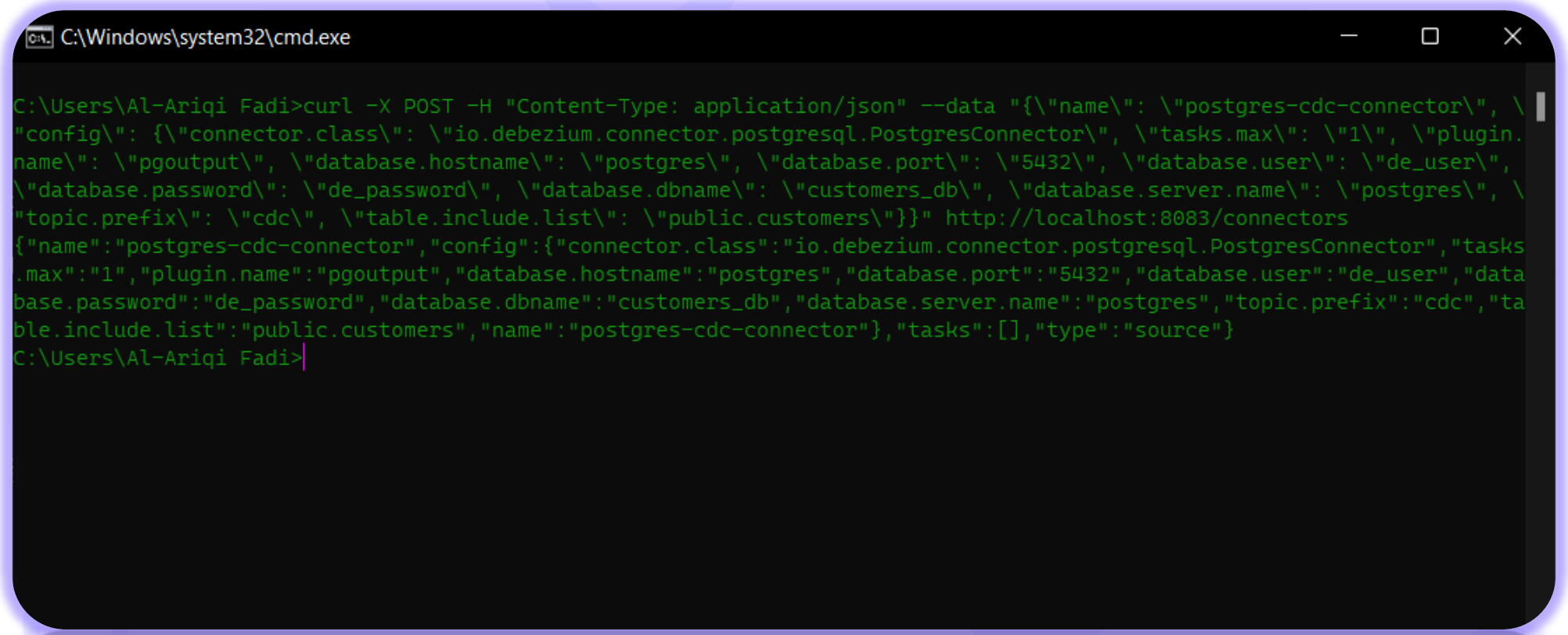
Running the Real-Time Ticket Generator



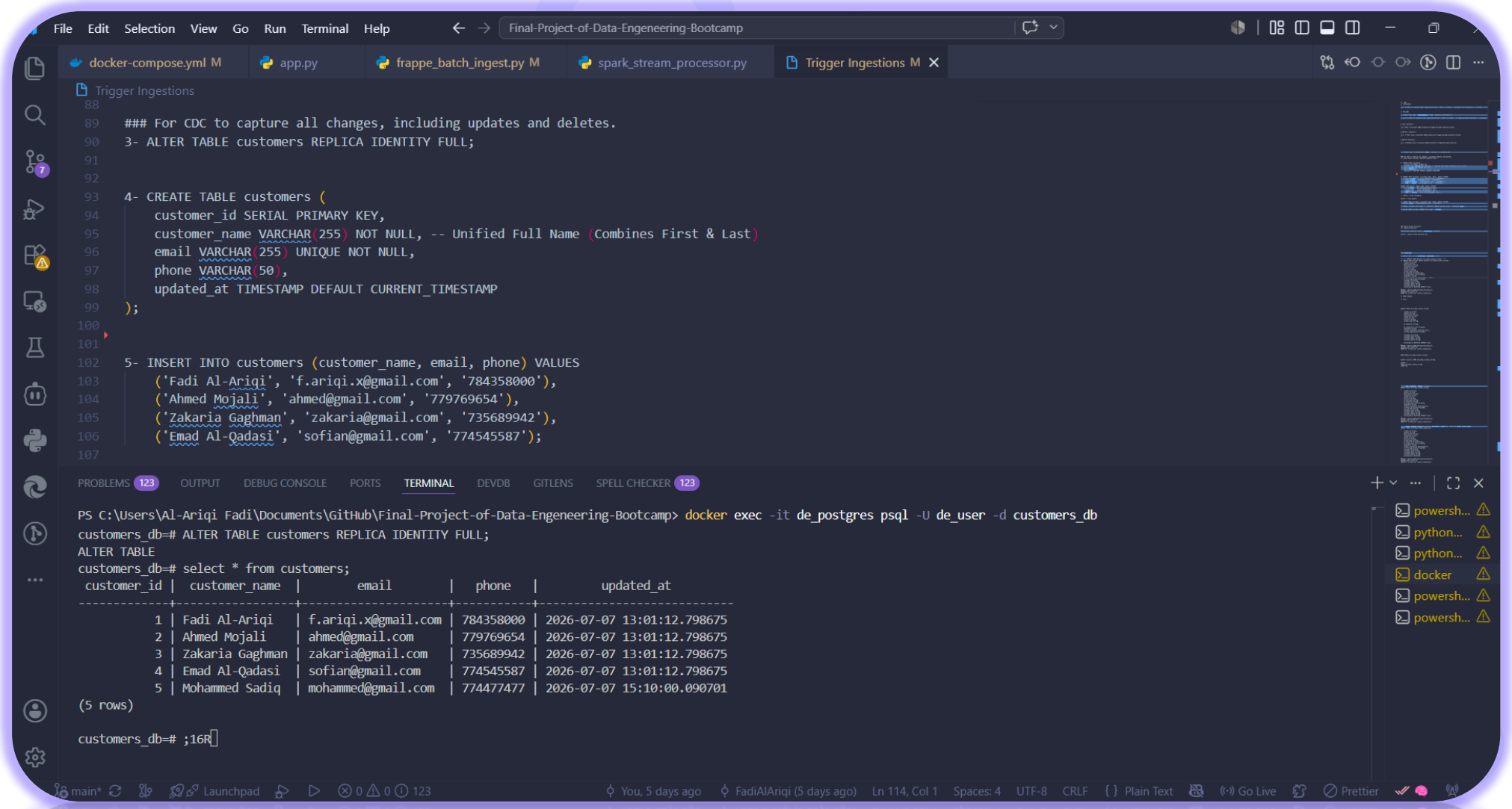
Running the Frappe Ticket Synchronization Service via API



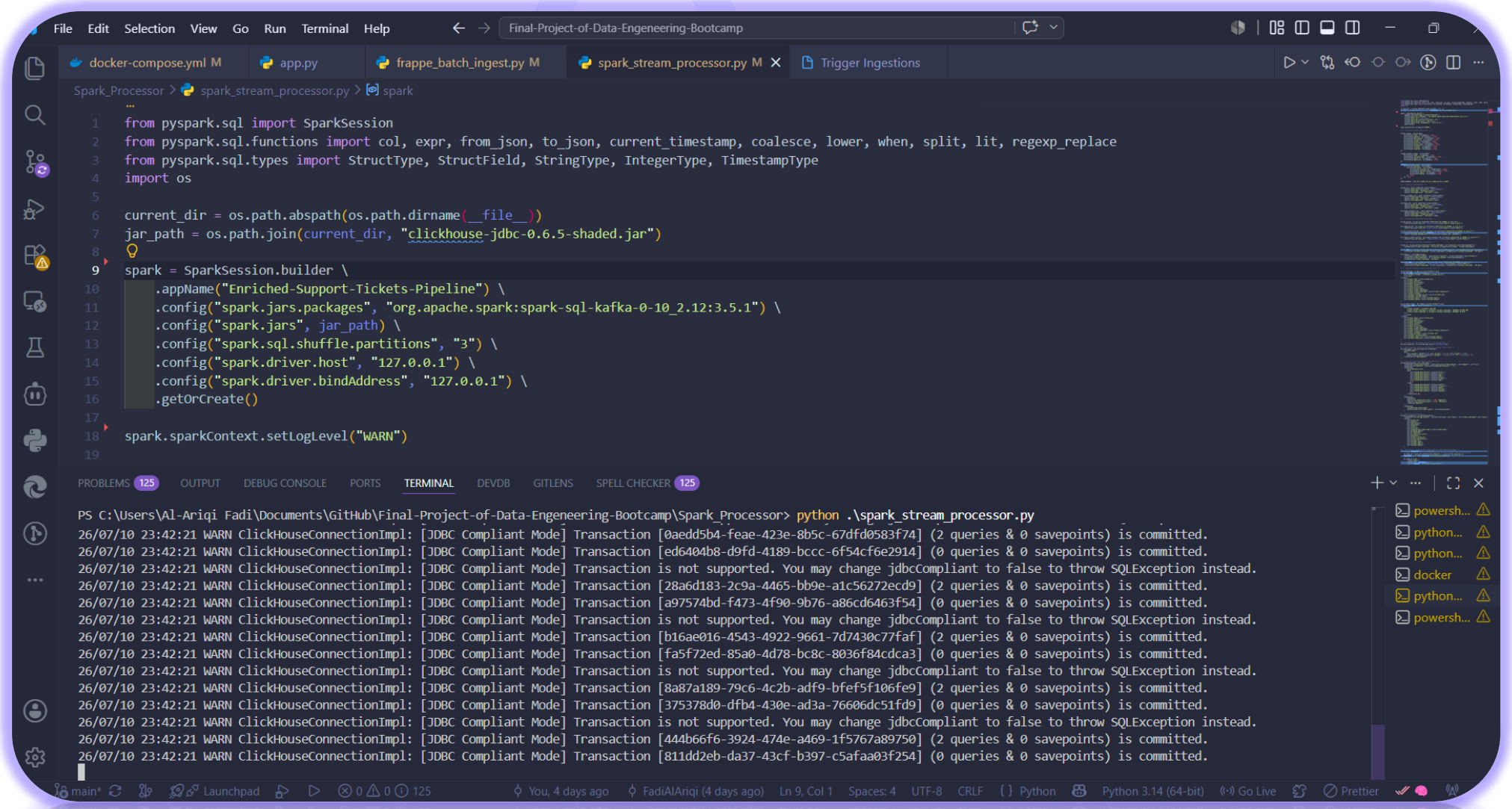
Creating the Debezium Connector for PostgreSQL Change Data Capture



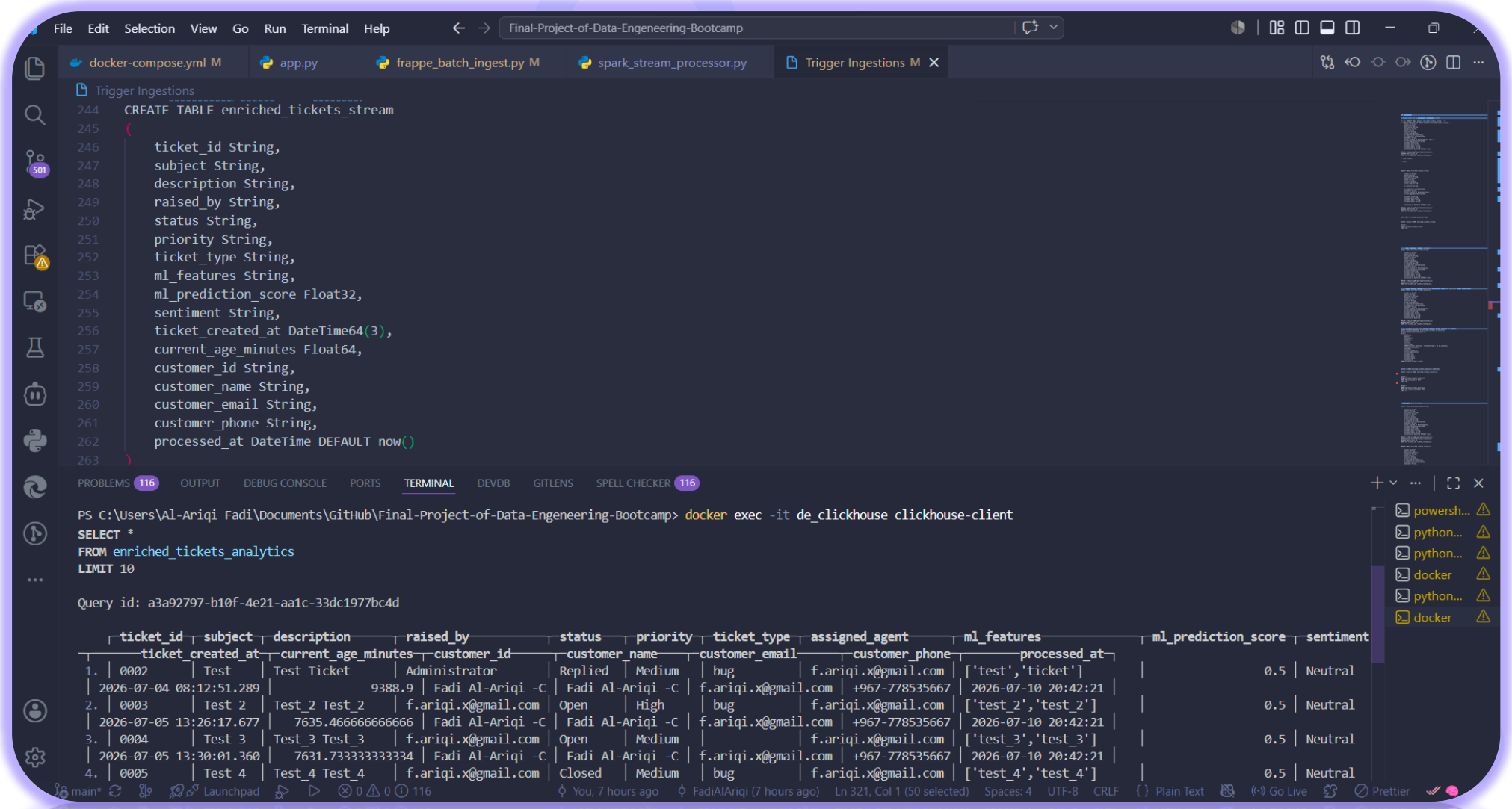
Creating Customer and Agent Data in PostgreSQL



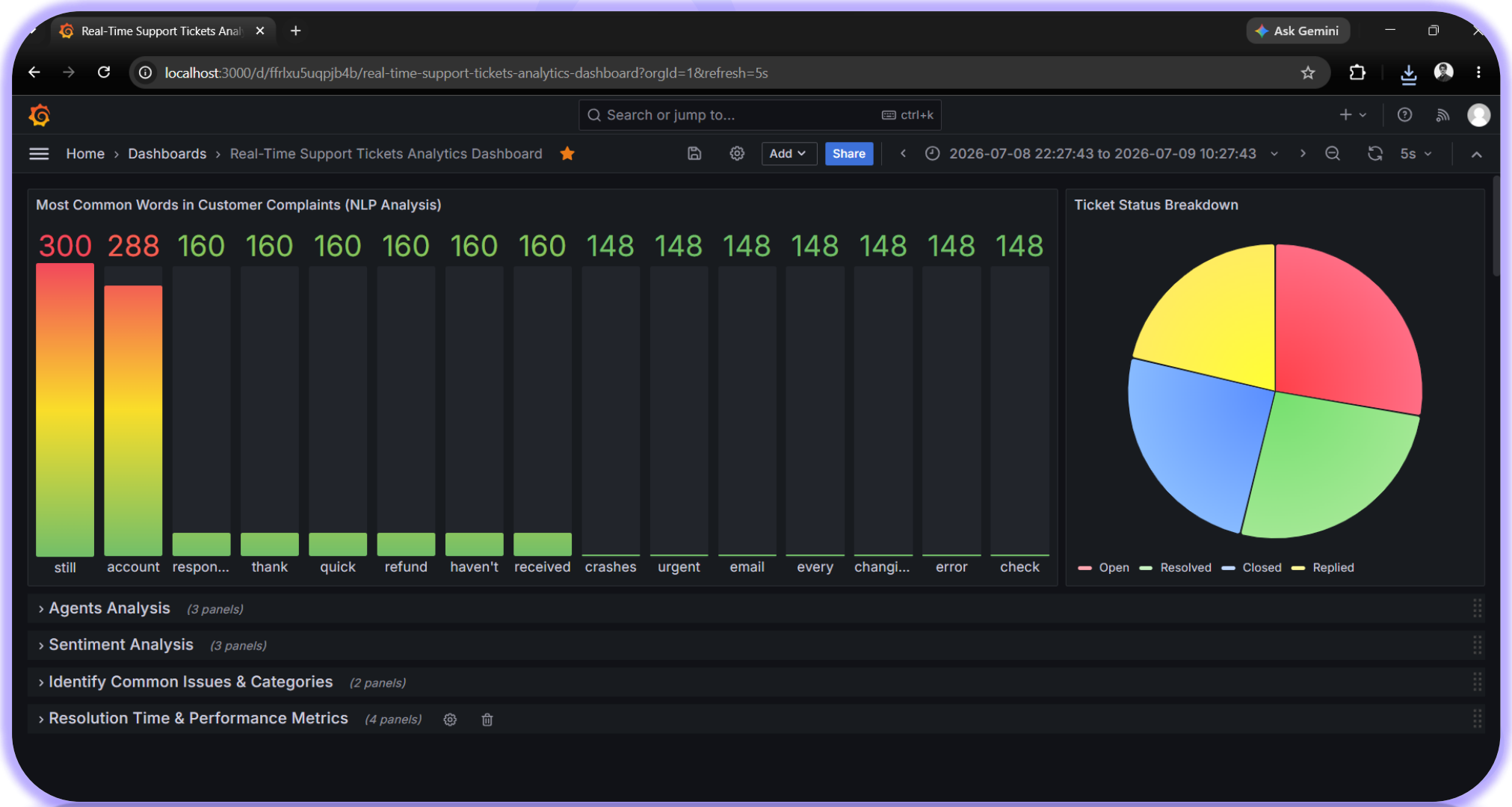
Running the Apache Spark Processing Pipeline



Processed Data Successfully Stored in ClickHouse



Grafana Analytics Dashboard – Overview



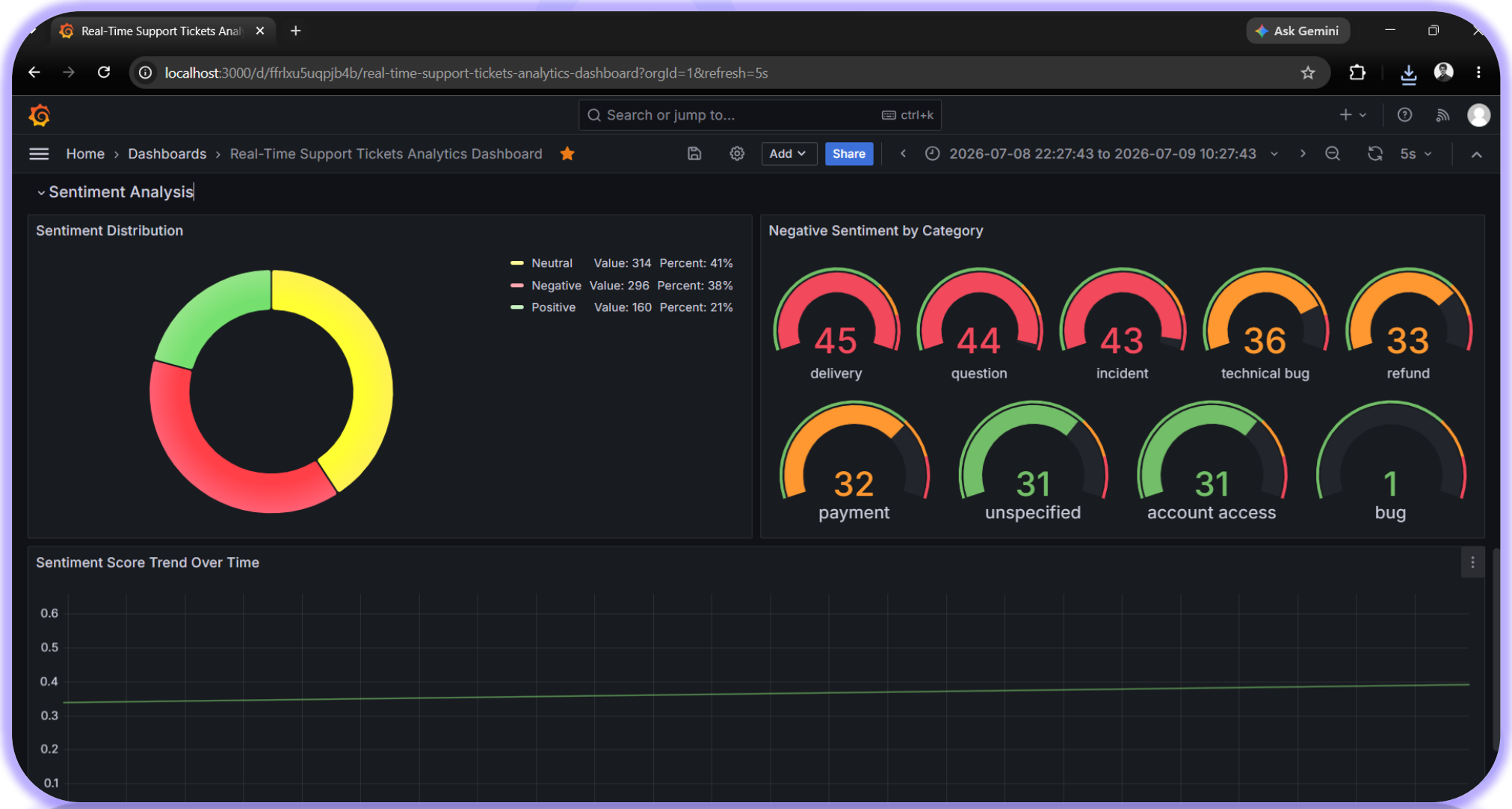
Grafana Dashboard – Agent Performance Analytics 1



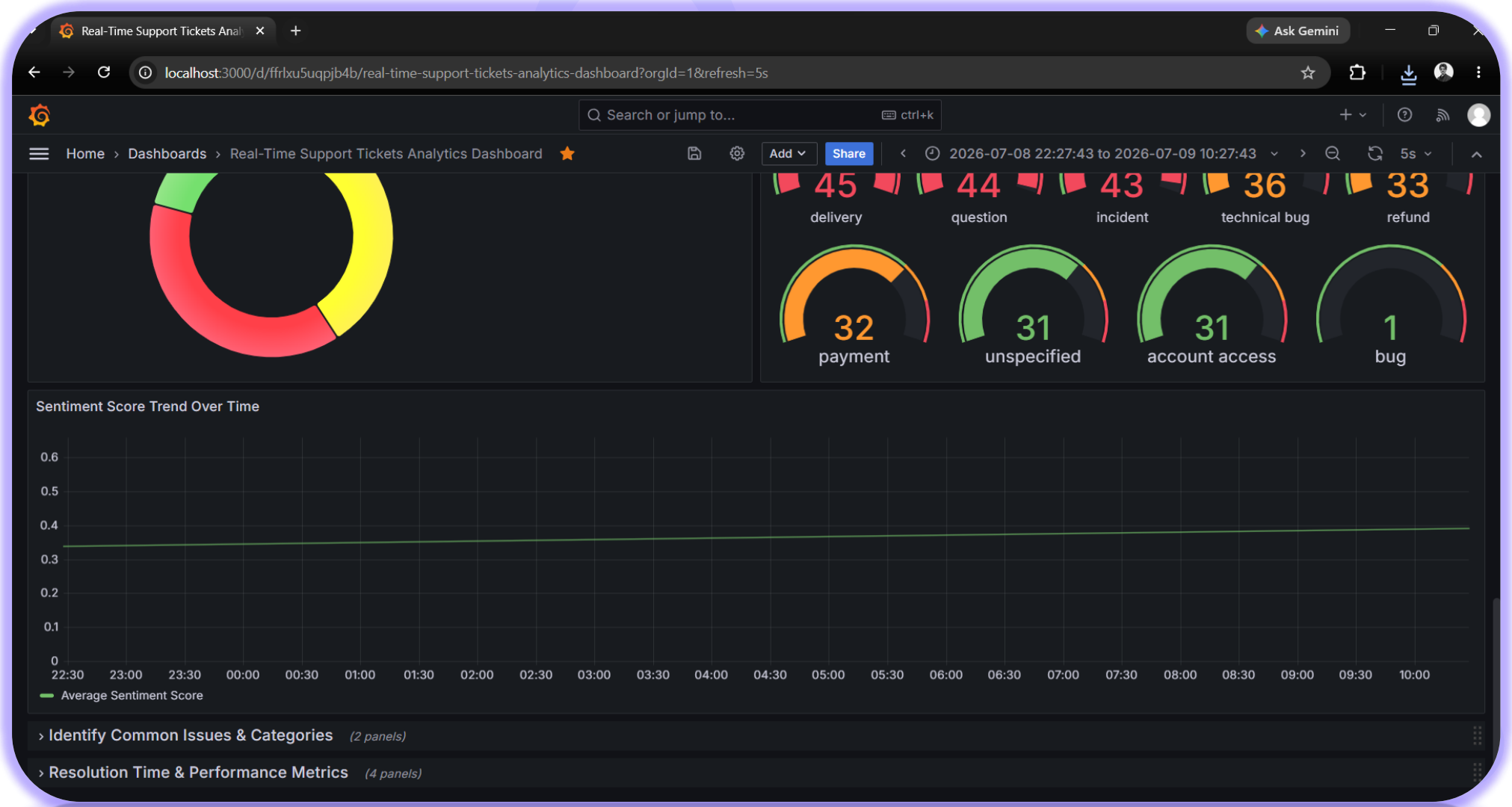
Grafana Dashboard – Agent Performance Analytics 2



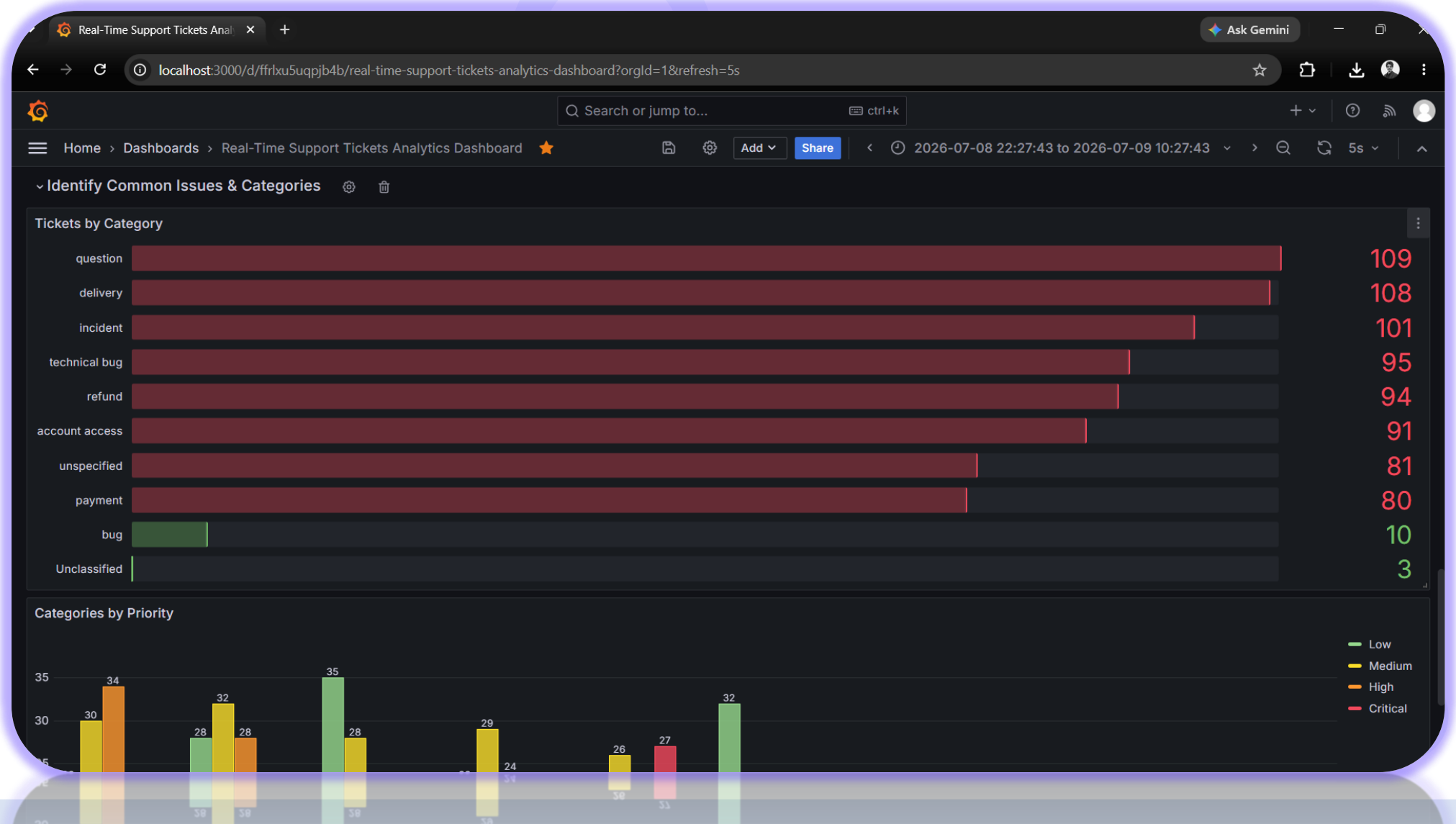
Grafana Dashboard – Customer Sentiment Analytics 1



Grafana Dashboard – Customer Sentiment Analytics 2



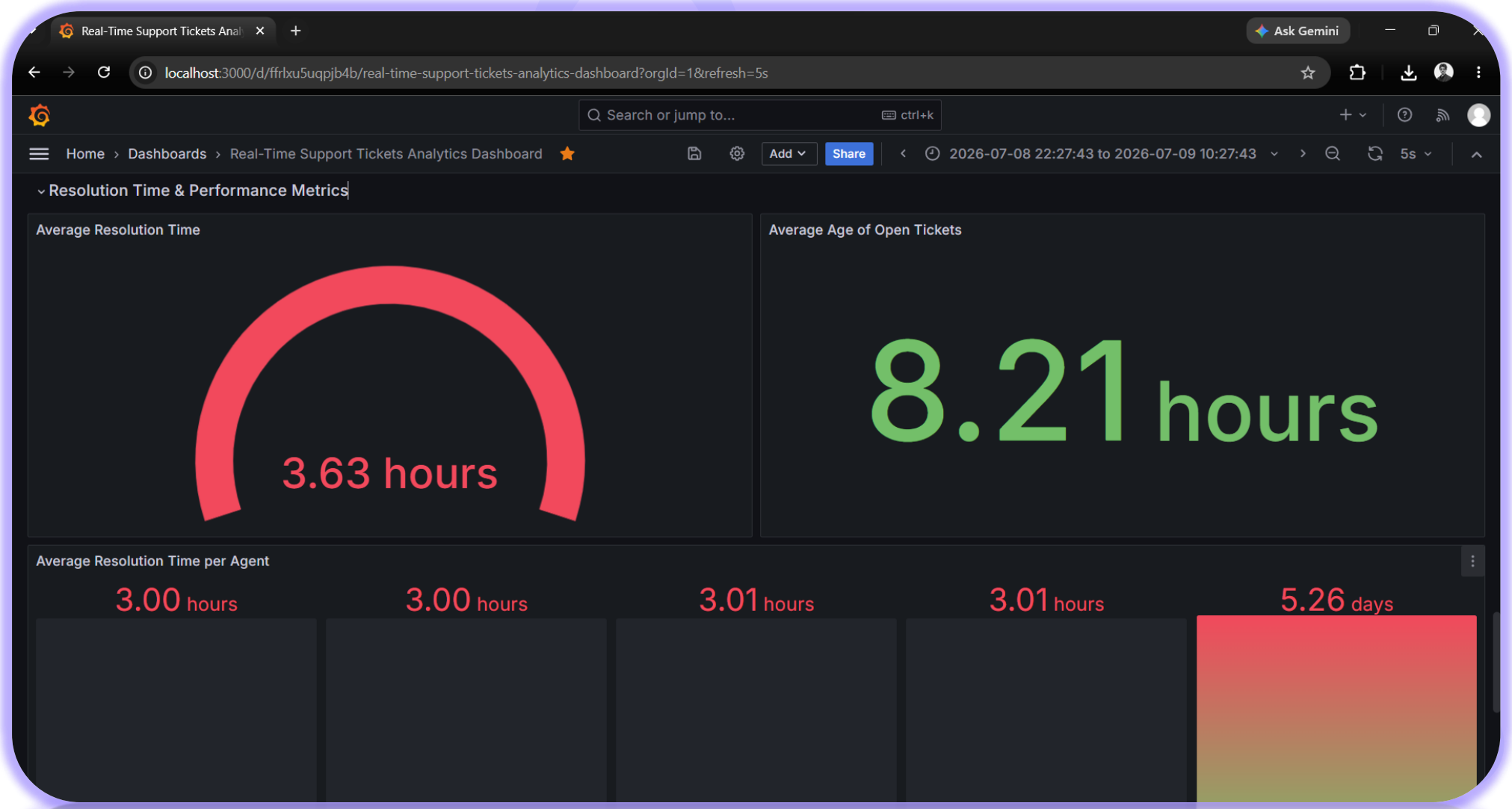
Grafana Dashboard – Common Issues & Category Analysis 1



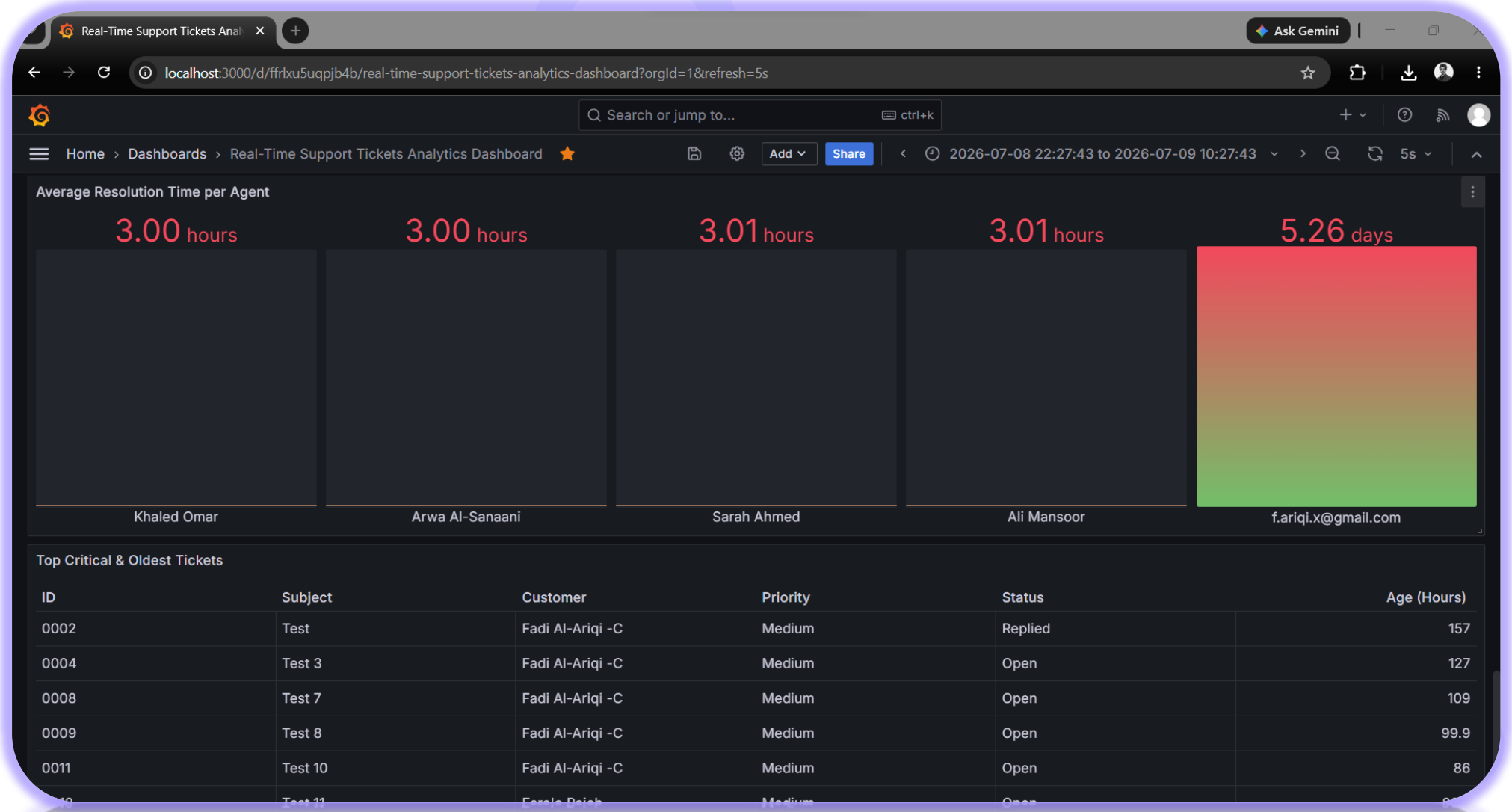
Grafana Dashboard – Common Issues & Category Analysis 2



Grafana Dashboard – Resolution Time & Performance Metrics 1



Grafana Dashboard – Resolution Time & Performance Metrics 2



Thank You for Your Attention

Eng. Fadi Al-Ariqi